

POLYARNY, Russian Federation

WMO: 243220

Lat: 66.400N

Lon: 112.030E

Elev: 1660

StdP: 13.84

Time Zone: 9.00 (E09)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-49.4	-44.1	-53.5	0.2	-49.6	-48.8	0.3	-44.5	14.6	-0.6	13.2	-4.8	5.5	290	0.804

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	80.2	60.7	76.0	59.1	71.8	57.2	62.8	75.7	60.6	71.6	58.6	69.3	7.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	77.1	66.2	55.7	70.2	64.3	53.7	65.1	63.4	29.1	76.0	27.6	72.0	26.2	69.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.7	12.2	DB	-50.8	85.1	6.4	3.7	-55.4	87.8	-59.1	90.0	-62.7	92.1	-67.4	94.8	(4)
(5)			WB	-50.8	67.1	6.4	2.8	-55.4	69.1	-59.1	70.7	-62.7	72.3	-67.3	74.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	-19.6	-19.2	0.9	18.1	34.8	55.1	59.4	52.4	37.7	15.0	-9.3	-19.8
	DBStd	31.36	14.27	12.72	14.03	11.99	10.45	9.58	8.60	7.96	8.73	11.92	12.58	13.87
	HDD50	12604	2158	1936	1523	956	485	55	17	64	380	1085	1780	2165
	HDD65	17473	2623	2356	1988	1406	935	319	218	398	820	1550	2230	2630
	CDD50	679	0	0	0	0	14	208	308	139	10	0	0	0
	CDD65	73	0	0	0	0	0	22	44	7	0	0	0	0
	CDH74	549	0	0	0	0	0	185	318	45	1	0	0	0
	CDH80	100	0	0	0	0	0	41	55	4	0	0	0	0
Wind	WSAvg	6.6	6.2	5.8	6.7	7.6	7.7	7.3	6.7	6.7	6.7	6.5	5.6	5.4
Precipitation	PrecAvg	14.5	0.5	0.4	0.6	0.8	1.4	2.0	2.0	2.3	1.5	1.4	0.9	0.6
	PrecMax	20.6	0.9	0.9	0.9	1.7	2.4	3.5	3.4	5.2	2.8	2.0	1.7	2.1
	PrecMin	10.7	0.1	0.1	0.2	0.3	0.5	0.7	0.5	0.6	0.2	0.5	0.4	0.2
	PrecStd	2.0	0.2	0.2	0.2	0.3	0.6	0.7	0.7	0.9	0.6	0.4	0.3	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	12.3	36.3	47.8	68.3	85.1	84.6	79.1	68.6	42.5	30.0	13.4
		MCWB	16.8	11.6	30.7	37.3	50.0	61.7	63.4	61.5	54.6	38.4	28.3	13.0
	2%	DB	8.5	7.4	30.3	41.7	61.2	78.9	80.8	73.2	57.5	37.6	21.5	8.4
		MCWB	8.1	6.8	26.2	33.4	46.9	59.1	61.3	57.7	47.7	35.0	21.1	7.9
	5%	DB	3.5	3.5	24.8	37.7	55.7	74.9	77.3	68.5	52.7	33.9	14.4	3.5
		MCWB	3.2	3.2	21.9	31.7	43.8	57.5	59.8	55.8	45.4	32.2	13.8	3.3
	10%	DB	-0.7	-0.8	19.9	34.1	50.4	70.1	73.6	64.6	48.8	30.1	7.1	-0.4
		MCWB	-1.0	-1.1	17.8	29.2	40.6	55.4	58.6	54.5	43.2	28.6	6.9	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.0	11.9	30.8	37.5	51.8	63.5	66.5	63.5	54.4	38.5	28.1	13.0
		MCDB	16.2	12.2	35.4	46.8	67.3	78.3	79.9	75.6	65.9	40.5	30.2	13.2
	2%	WB	8.0	6.8	25.9	34.5	47.7	61.4	63.9	60.1	49.4	35.3	20.6	7.9
		MCDB	8.4	7.3	29.9	40.1	60.2	74.7	76.8	68.9	54.6	37.0	21.2	8.4
	5%	WB	3.3	3.2	22.1	32.1	44.5	59.3	61.8	57.6	46.4	32.1	13.7	3.3
		MCDB	3.6	3.4	24.9	37.3	54.7	71.1	73.2	66.4	52.0	33.9	14.1	3.5
	10%	WB	-1.5	-1.6	18.1	29.5	41.1	57.2	59.6	55.3	43.9	28.5	6.9	-0.7
		MCDB	-1.2	-1.4	20.1	34.1	49.9	68.9	70.6	63.5	47.7	29.6	7.3	-0.5
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.6	13.8	14.7	13.3	16.2	15.7	13.6	10.8	8.8	9.6	9.4
		MCDBR	13.0	12.9	14.7	15.1	18.5	20.0	19.4	18.8	15.6	10.4	16.8	11.6
		MCWBR	12.9	12.8	12.1	10.8	10.2	8.8	8.7	8.8	9.3	8.8	16.5	11.5
	5% WB	MCDBR	13.1	13.0	14.4	14.5	17.7	18.1	17.9	16.5	14.4	10.0	17.0	11.6
		MCWBR	13.0	12.9	12.1	11.3	10.5	9.4	9.5	9.4	9.6	8.7	16.6	11.5
Clear-Sky Solar Irradiance	taub	0.181	0.232	0.277	0.305	0.336	0.358	0.408	0.401	0.299	0.241	0.171	0.174	
	taud	1.863	2.055	2.087	2.115	2.239	2.406	2.247	2.212	2.485	2.196	1.888	1.874	
	Ebn at Noon	70	198	245	269	273	270	248	234	246	194	70	7	
	Edh at Noon	5	17	29	36	36	32	36	34	20	15	4	1	
All-Sky Solar Radiation	RadAvg	26	248	787	1448	1757	1714	1591	1067	603	263	57	3	
	RadStd	3	16	42	74	107	158	128	136	67	30	6	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page